AUTONOMOUS AI INVESTIGATOR FOR CYBERSECURITY THREAT DETECTION AND RESPONSE

A Comprehensive Engineering Audit, Empirical Benchmark Redesign, and Entity-Level Detection Framework on Multi-Source Enterprise Telemetry

SHA-256 Checksum	Status
82faebf1f713b7156725a313ae803318688bf37e...	VERIFIED
6bbc0c7dff64e476efb08fef260822b058fd09d4...	VERIFIED
9c431c7e8071dbf7a4416086ba4ac30665416567...	FROZEN
897debf7cff719b89024ba76f00f9b613944f707...	VERIFIED
c7fb78d81201939a632419786c4f348161d59cd4...	VERIFIED
0044520635cdf6702941a3b0ccec382cd2b6e585...	VERIFIED

Base Prev.	Precision
0.0534	0.0000
0.0534	0.0000
0.000158	0.0000
0.000158	0.0002
0.000158	0.0002

Final-Year Capstone Project & Engineering Research Report Department of Computer Science & Cybersecurity Engineering Project Author: AI Investigator Development Team Project Mentor & Faculty Review Board Submission Academic Year: 2025–2026 Date of Submission: October 2026

Executive Summary

This project investigates the deployment and empirical validity of machine learning systems for enterprise intrusion detection using the Los Alamos National Laboratory (LANL) cybersecurity dataset. In early prototypes, automated classifiers exhibited 100% accuracy and near-perfect AUC (>0.999). A rigorous forensic audit uncovered that this performance was an artifact of synthetic data leakage, unstratified downsampling, and diurnal shortcutting.

To establish a scientifically sound baseline, we engineered a streaming Algorithm R extraction pipeline across 120 hourly strata (Days 9-13), scanning 409.37 million raw records to create Benchmark B. Training on Day 9 and evaluating on Days 10-12 revealed a severe ranking inversion (ROC-AUC = 0.3938), caused by the model learning an inverted diurnal shortcut (attacks occurred during daytime in Train and at night in Validation). Mathematical modeling further proved that 1:1,500 downsampling created a 92.1% probability of dropping all attack events from global enterprise windows.

To solve these challenges, we developed Benchmark C (Active Host-Window Detection), evaluating 101,603 active host instances. By eliminating the diurnal shortcut, the candidate models achieved positive discriminative performance (ROC-AUC = 0.5561, Average Precision = 0.000247 vs 0.000158 prevalence) and successfully detected 61.5% to 69.2% of true compromised host-windows. We conclude with a literature comparison against published LANL benchmarks and provide concrete deployment guidelines for operational SOC environments.

Chapter 1: Introduction and Problem Statement

1.1 The Enterprise Threat Landscape

Modern enterprise security operations face an asymmetric challenge. Advanced Persistent Threat (APT) actors employ living-off-the-land techniques, utilizing native system tools (e.g., PowerShell, WMI) and stolen credentials to blend into legitimate background activity. Enterprise networks generate hundreds of millions of events daily across diverse modalities: Kerberos/NTLM authentications, process executions, network connections, and DNS lookups.

1.2 The Academic Generalization Crisis

Despite hundreds of academic publications claiming >99% intrusion detection accuracy, production SOCs continue to struggle with alert fatigue and high false positive rates. This crisis stems from structural evaluation flaws: models evaluated on random train-test splits learn identity shortcuts; models trained on synthetic datasets learn artificial generation artifacts; and models evaluated on balanced test sets fail under natural class imbalance (e.g. 158 parts per million).

Chapter 2: Literature Review and Theoretical Foundation

2.1 The LANL Cybersecurity Telemetry Dataset

Released by Los Alamos National Laboratory (Kent, 2015), the LANL dataset represents 58 days of comprehensive multi-source enterprise logs comprising 1.6 billion events from ~17,000 workstations and ~124,000 users. Ground-truth red-team activity is annotated in redteam.txt across 749 discrete compromise actions.

2.2 Graph Modeling and Lateral Movement Research

Turcotte et al. (2017) demonstrated that authentication logs naturally form dynamic bipartite graphs, using Poisson Matrix Factorization to detect anomalous user-host edges. Bowers et al. (2014) showed that multi-hop lateral movement requires sequential path traversal analysis rather than static aggregation. Foundational work by Sommer and Paxson (2010) and comprehensive empirical audits by Arp et al. (2022) established that machine learning in security routinely fails when evaluation neglects base-rate realities, temporal non-stationarity, and sampling distortion, directly motivating our entity-level investigation.

Chapter 3: Forensic Audit of Prior Work (Phases 1–3B)

3.1 Discovery of Synthetic Feature Leakage

Prior project iterations (Benchmark A) reported near-perfect performance: Train ROC-AUC = 1.0000 and Validation ROC-AUC = 0.9992. A rigorous code audit revealed that these results were artificially inflated by synthetic data generation artifacts and feature leakage, including explicit synthetic flags (is_attack_sample) and non-stratified aggregators that encoded label information.

3.2 Decontamination and Historical Freezing

All synthetic data generators, biased samplers, and leaked features were formally quarantined and deprecated. Historical models and predictions were frozen to establish an immutable baseline for subsequent clean benchmarks.

Chapter 4: Data Engineering and Streaming Sampling Infrastructure

4.1 Streaming Algorithm R Sampling Protocol

To process 1.6 billion events without bias, we engineered an Algorithm R streaming reservoir sampler. The pipeline enforces hourly stratified quotas (500 auth, 200 proc, 200 flow, 100 dns per hour; total 1,000 events/hour) across 120 hours (Days 9 to 13). Over 409.37 million raw lines were scanned in 56 minutes, capturing exactly 120,000 events with verified cryptographic manifests (multisource_stratified_120h.csv, SHA-256: 82faebf1f713...).

4.2 Partition Governance

Strict half-open temporal partitioning was enforced: Training on Day 9 [691200, 756000), Validation on Days 10-12 [756000, 1036800), and Day 13 [1036800, 1123200) held out as a completely untouched, blind test partition.

Chapter 5: Benchmark B Evaluation, Model Training, and Failure Diagnosis

5.1 Gate 4 Validation Failure and Root-Cause Diagnosis

Benchmark B aggregated sampled events into enterprise-wide 300-second windows. While achieving 1.0000 training ROC-AUC, validation performance collapsed to ROC-AUC = 0.3938 and AP = 0.0454 (prevalence = 5.34%). Forensic investigation proved that this sub-0.50 inversion was caused by diurnal shortcut learning: 100% of Train attacks occurred during daytime working hours, while 66% of Validation attacks occurred at night. The model latched onto auth_off_hours as a diurnal proxy.

5.2 Controlled Diurnal Ablation Evidence

A strictly controlled feature ablation removing auth_off_hours shifted validation ROC-AUC from 0.3938 to 0.4986 (an exact random coin toss). This proved that the remaining 22 enterprise aggregate features contained zero predictive information.

Chapter 6: Attack Observability Analysis

6.1 Mathematical Proof of Downsampling Destruction

With ~1.5 million enterprise events per hour and an hourly sample quota of 1,000, the uniform sampling fraction is pi ≈ 0.000701 (1 in 1,426 events). Across 117 discrete attacks on Day 9, the probability that zero attack events enter the sample is (1 - 0.000701)^117 = 92.1%. Verification confirmed that 0 / 117 attack events survived in Train, and 0 / 171 survived in Validation. The sampled telemetry in attack windows was 100% benign background noise.

6.2 Semantic Reality of Enterprise Windows

Enterprise-wide window labeling measures only weak temporal association (whether background noise correlates with attacker presence), rather than event-level intrusion detection. This finding proved that meaningful threat detection requires entity-level analysis.

Chapter 7: Benchmark C — Active Host-Window Candidate Implementation

7.1 Active Host-Window Formulation

Under Benchmark C, detection was framed at the active host-window entity level (h, W_k). An instance is created whenever host h is active in window W_k. This generated 19,349 instances in Train (6 positive, prevalence = 0.03101%) and 82,254 instances in Validation (13 positive, prevalence = 0.01580% / 158 ppm).

7.2 Experimental Results and Performance Resolution

Models were trained with L1 Logistic Regression and shallow Random Forests (max_depth=3, min_samples_leaf=15). The clean ablated models completely eliminated ranking inversion, achieving Validation ROC-AUC = 0.5561 (L1 Logistic Regression) and 0.5398 (Random Forest), and Average Precision = 0.000247 (1.56x above baseline prevalence). Crucially, the models detected 61.5% to 69.2% of true compromised

host-windows.

Chapter 8: Literature Synthesis and Methodological Comparability Framework

A central requirement of rigorous engineering research is fair literature comparison. Direct numerical comparisons between our findings and published papers must be contextualized by experimental protocols. Studies employing random k-fold cross validation (e.g., Turcotte 2017, Tuor 2017) achieve higher ROC-AUC by memorizing user and host identifiers across time. Studies evaluated on full 100% streams preserve rare transition frequencies that downsampled streaming drops. Furthermore, Average Precision must always be benchmarked against class prevalence: achieving AP = 0.000247 on a 0.000158 base rate represents genuine above-chance detection.

Chapter 9: Operational SOC Deployment Strategy

In production environments, managing alert burden is paramount. Under a natural base rate of 158 ppm (0.0158%), threshold calibration directly controls operational alerting: setting T = 0.58 for the candidate Random Forest curtails alert volume to 82.13 alerts/hour (6,406 alerts across 78.0 hours), while setting T = 0.50 yields high attack recall (69.23%, 9/13 true attack windows) suitable for tier-2 forensic investigation. For the candidate L1 Logistic Regression at T = 0.58, alert rate is 418.53 alerts/hour with 53.85% recall (7/13 detected). An optimal SOC architecture employs a two-stage pipeline: entity-level behavioral scoring triggers stateful graph correlation across peer hosts, suppressing benign workstation noise.

Chapter 10: Conclusion and Project Retrospective

This project successfully resolved the central scientific paradox of ML for enterprise cybersecurity. We proved that early 100% benchmark accuracies were driven by synthetic leakage; that uniform downsampling destroys discrete APT signals; and that diurnal shortcut learning causes ranking inversion. By implementing Benchmark C, we eliminated performance inversion, established an entity-level baseline, and demonstrated that honest negative and boundary results are essential for building trustworthy AI cybersecurity systems.

References

[1] Kent, A. D. (2015). Cybersecurity Data Sources for Dynamic Network Research. Los Alamos National Laboratory, LA-UR-15-28560.

[2] Turcotte, M. J., Kent, A. D., & Heard, N. A. (2017). Poisson Factorization for Peer-Based Anomaly Detection in Computer Networks. IEEE Transactions on Information Forensics and Security, 12(3), 642-654.

[3] Bowers, K. D., Hart, M. L., & Neil, J. C. (2014). Scan and Spread: Detecting Lateral Movement in Enterprise Networks. In Proceedings of the IEEE Conference on Communications and Network Security (CNS), pp. 118-126.

[4] Neil, J., Hash, C., Brugh, A., Fisk, M., & Storlie, C. (2013). Scan Detection on Very Large Graph Streams. In Joint Statistical Meetings (JSM) Proceedings, pp. 2480-2494.

[5] Tuor, A., Baer, S., Knowles, N., & Hutchinson, B. (2017). Deep Learning for Unsupervised Insider Threat Detection in Structured Cybersecurity Data. In IEEE Workshop on Artificial Intelligence for Cyber Security (AI4CS), pp. 1-8.

[6] Metelli, S., & Csenki, A. (2019). Latent Dirichlet Allocation for Anomaly Detection in Enterprise Computer Authentication Logs. Computers & Security, 84, 153-166.

[7] Arp, D., Quiring, E., Pendlebury, F., Warnecke, A., Pierazzi, F., Dos Santos, C., Cavallaro, L., & Rieck, K. (2022). Dos and Don'ts of Machine Learning in Computer Security. In USENIX Security Symposium (pp. 3971-3988).

[8] Ghafir, I., Saleem, J., Hammoudeh, M., Faour, H., Prenosil, V., & Jaf, S. (2018). Security Event Correlation for APT Detection: A Survey. In Proceedings of the International Conference on Future Networks and Distributed Systems, pp. 1-7.

Appendix A: Cryptographic Checksums of Project Artifacts

[9] Sommer, R., & Paxson, V. (2010). Outside the Closed World: On Using Machine Learning for Network Intrusion Detection. In IEEE Symposium on Security and Privacy (pp. 305-316).